

Corresponding author address

May 16, 2026

Editors-in-Chief  
SIAM Journal on Financial Mathematics

Dear Editors,

Please consider the manuscript “Structural Ambiguity Amplification in Near-Critical Hawkes Market Making” for publication as a full-length research article in *SIAM Journal on Financial Mathematics*.

The manuscript studies structural uncertainty in Hawkes-driven market making. Its central contribution is a reproducible mathematical and computational package showing how ambiguity in the branching matrix is amplified by the Hawkes resolvent near criticality. The paper distinguishes relative critical-slack ambiguity from absolute branching-matrix ambiguity, verifies a finite-state robust Bellman layer and a multitype Hawkes HJBI/control bridge for the stated exponential model class, and evaluates the resulting policy diagnostics on simulated Hawkes paths, public LOBSTER samples, Binance aggregate-trade events, and crypto level-2 depth data.

The accompanying reproducibility package contains source code, tests, experiment scripts, public-data diagnostics, SOTA comparison tables, reference audits, and a strict submission validator. The current validator checks artifact presence, TeX logs, citation integrity, SOTA metadata, displayed-depth queue replay diagnostics, M-series benchmark controls, and proof/scope audit anchors.

To the best of our knowledge, this manuscript is original, is not under review elsewhere, and has not been published previously.

Artificial intelligence disclosure draft: portions of the manuscript drafting, code scaffolding, experiment automation, documentation, and validation scripts were prepared with assistance from OpenAI Codex, a GPT-5 based coding assistant, on May 16, 2026. The human authors are responsible for all mathematical claims, proofs, citations, code, data processing, numerical experiments, and final text included in the submitted manuscript.

Thank you for your consideration.

Sincerely,

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